Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI				**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**

EXPERIMENT:- 4

Aim: To set up a Continuous Integration Pipeline using Jenkins/GitHub Act.

Lab Outcome: Mapped LO3

LO3: Build and manage automated CI/CD pipelines for application development, testing, packaging and deployment.

Theory:

- Continuous Integration (CI) & Continuous Delivery (CD)

  - Continuous Integration (CI): A modern DevOps development practice where developers merge their code changes frequently into a central shared repository (such as GitHub). Every commit triggers an automated build and execution of tests.
    - Primary Objective: Detect integration bugs and software defects early in the software development lifecycle (SDLC), preventing "integration hell" when multiple developers work simultaneously.
  - Continuous Delivery / Continuous Deployment (CD): An extension of CI where code changes are automatically prepared (Delivery) or deployed directly (Deployment) to testing or production environments once all continuous integration builds pass successfully.

- Overview of Jenkins

  - Jenkins Server: An open-source automation server written in Java that serves as the foundation for CI/CD pipelines.
  - Core Responsibilities:
    - Automation: Automates repetitive engineering tasks, including code compilation, unit testing, static code analysis, artifact generation, and deployment.

Name:- HITEN PAHILAJANI				**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**

- Plugin Ecosystem: Supports thousands of plugins to integrate seamlessly with version control systems (Git, SVN), build tools (Maven, Gradle, npm), and deployment platforms (Docker, Kubernetes, AWS).
- Event Triggers: Can trigger workflows automatically via Git webhooks, scheduled cron jobs, or manually through its Web User Interface (UI).

● Jenkins Architecture

- Jenkins Master (Controller): Handles configuration, schedules jobs, monitors build executors, dispatches builds to agents, and serves the Web UI.
- Jenkins Agents (Nodes/Workers): External machines or containers that execute the actual build steps dispatched by the Controller, allowing distributed execution and scaling.

● Types of Jenkins Pipelines

A Jenkins Pipeline is a suite of plugins that supports implementing and integrating continuous delivery pipelines into Jenkins using code (often stored in a Jenkinsfile). There are two main types of pipeline syntaxes:

A. Declarative Pipeline

- Syntax Format: Enclosed inside a pipeline {} block.
- Characteristics:
  - Strict, predefined structure designed to be simple, highly readable, and maintainable for standard CI/CD workflows.
  - Offers superior syntax validation and error checking directly within the Jenkins UI prior to execution.
  - Organized strictly into defined sections such as agent, stages, stage, and steps.

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI				**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**

B. Scripted Pipeline

- Syntax Format: Enclosed inside a node {} block.
- Characteristics:
  - Built using full Apache Groovy scripting capability, offering maximum procedural flexibility and control.
  - Ideal for advanced, dynamic scenarios requiring complex control loops, conditional execution, or specialized logic.
  - Requires a higher learning curve due to its unstructured, code-heavy nature.

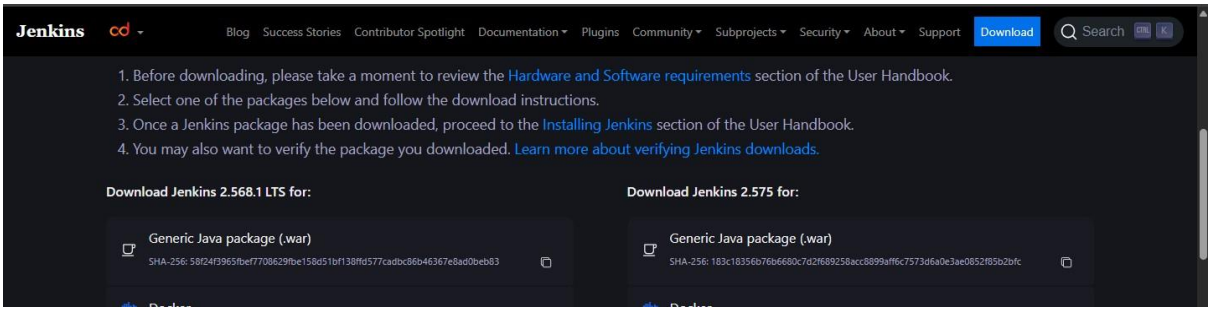
Execution:

Step 1: Java (JDK) Installation Verification

```
C:\Users\LENOVO>java -version
java version "22.0.2" 2024-07-16
Java(TM) SE Runtime Environment (build 22.0.2+9-70)
Java HotSpot(TM) 64-Bit Server VM (build 22.0.2+9-70, mixed mode, sharing)
```

Step 2: Jenkins Server Launch

Downloaded the Generic Java package (jenkins.war) from the official website.

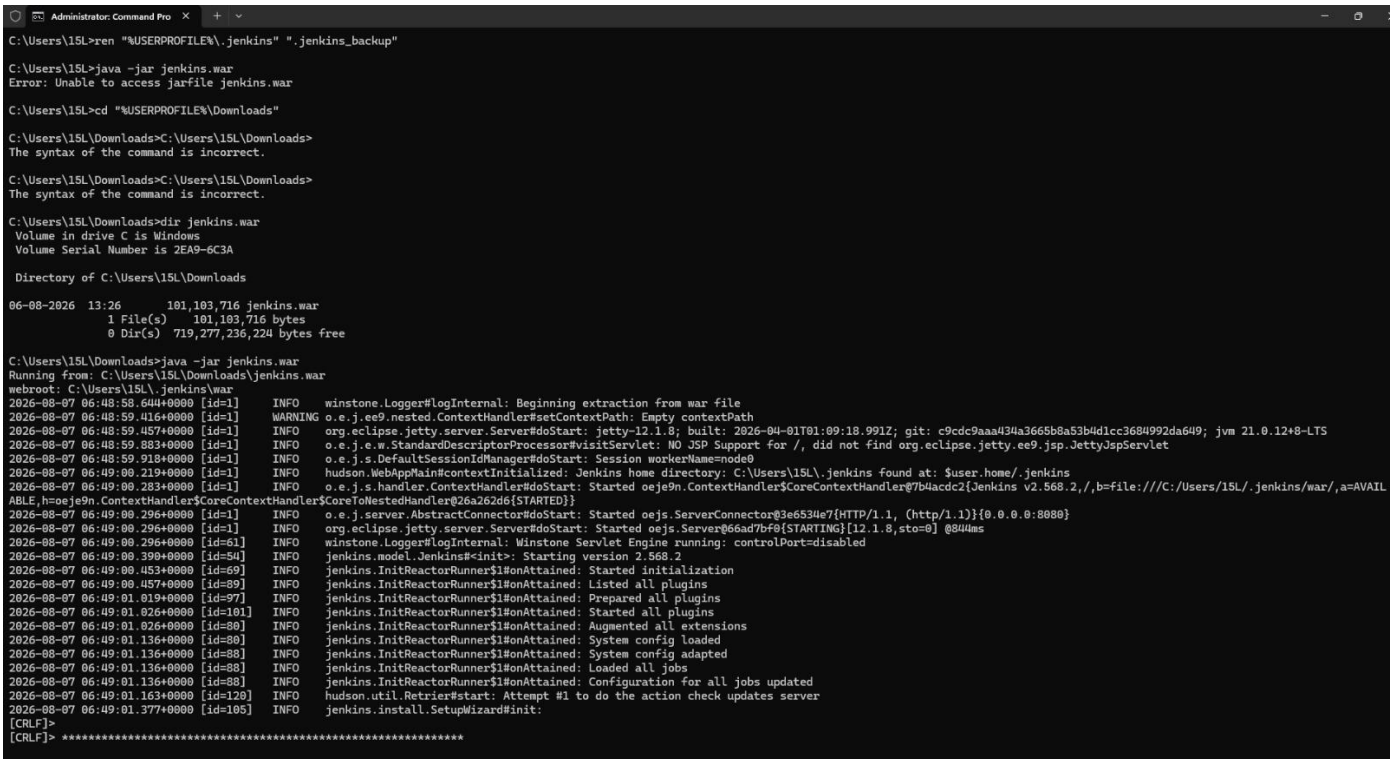


Executed Jenkins using the --enable-future-java flag to support JDK 22:

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI				**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**



```
Administrator: Command Pro
C:\Users\15L>ren "%USERPROFILE%\.jenkins" ".jenkins_backup"

C:\Users\15L>java -jar jenkins.war
Error: Unable to access jarfile jenkins.war

C:\Users\15L>cd "%USERPROFILE%\Downloads"

C:\Users\15L\Downloads>C:\Users\15L\Downloads>
The syntax of the command is incorrect.

C:\Users\15L\Downloads>C:\Users\15L\Downloads>
The syntax of the command is incorrect.

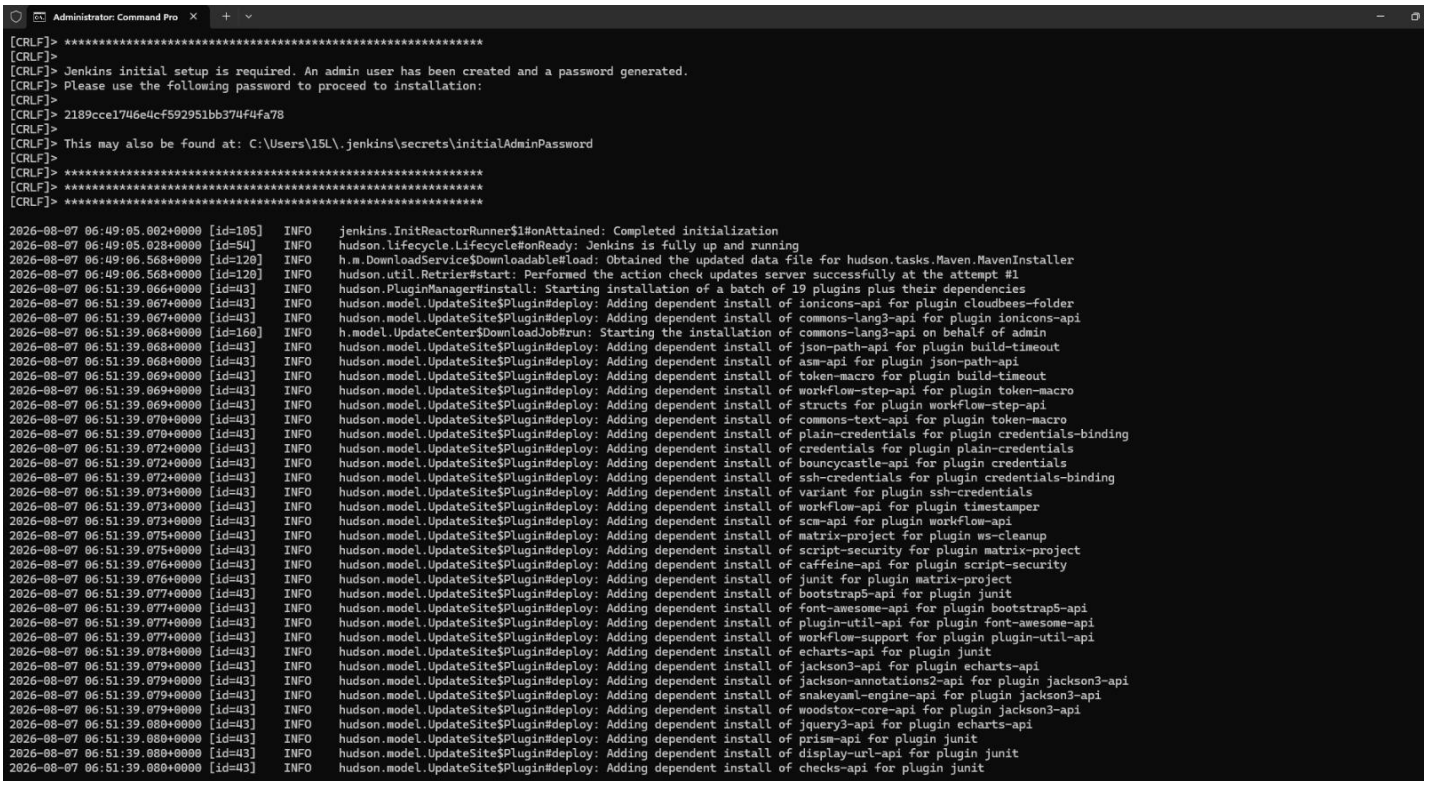
C:\Users\15L\Downloads>dir jenkins.war
 Volume in drive C is Windows
 Volume Serial Number is 2EA9-6C3A

 Directory of C:\Users\15L\Downloads

06-08-2026  13:26       101,103,716 jenkins.war
               1 File(s)    101,103,716 bytes
               0 Dir(s)  719,277,236,224 bytes free

C:\Users\15L\Downloads>java -jar jenkins.war
Running from: C:\Users\15L\Downloads\jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-07 06:48:58.644+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-07 06:48:59.416+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-07 06:48:59.457+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09:18.991Z; git: c9cdc9aaa434a3665b8a53b4d1cc3684992da649; jvm 21.0.12+8-LTS
2026-08-07 06:48:59.883+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-07 06:48:59.918+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-08-07 06:49:00.219+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\15L\.jenkins found at: $user.home/.jenkins
2026-08-07 06:49:00.283+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@7b4acdc2{Jenkins v2.568.2,/,b=file:///C:/Users/15L/.jenkins/war/,a=AVAIL
ABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@26a262d6{STARTED}}
2026-08-07 06:49:00.296+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@3e6534e7{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-08-07 06:49:00.296+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@66ad7bf0{STARTING}[12.1.8,sto=0] @844ms
2026-08-07 06:49:00.296+0000 [id=61]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-07 06:49:00.390+0000 [id=54]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.2
2026-08-07 06:49:00.453+0000 [id=69]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-07 06:49:00.457+0000 [id=89]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-07 06:49:01.019+0000 [id=97]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-07 06:49:01.026+0000 [id=101]   INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-07 06:49:01.026+0000 [id=80]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-07 06:49:01.136+0000 [id=80]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-07 06:49:01.136+0000 [id=88]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-07 06:49:01.136+0000 [id=88]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-07 06:49:01.136+0000 [id=88]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-07 06:49:01.163+0000 [id=120]   INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-08-07 06:49:01.377+0000 [id=105]   INFO    jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> ************************************************************
```

Step 3: Web Setup & Admin Configuration



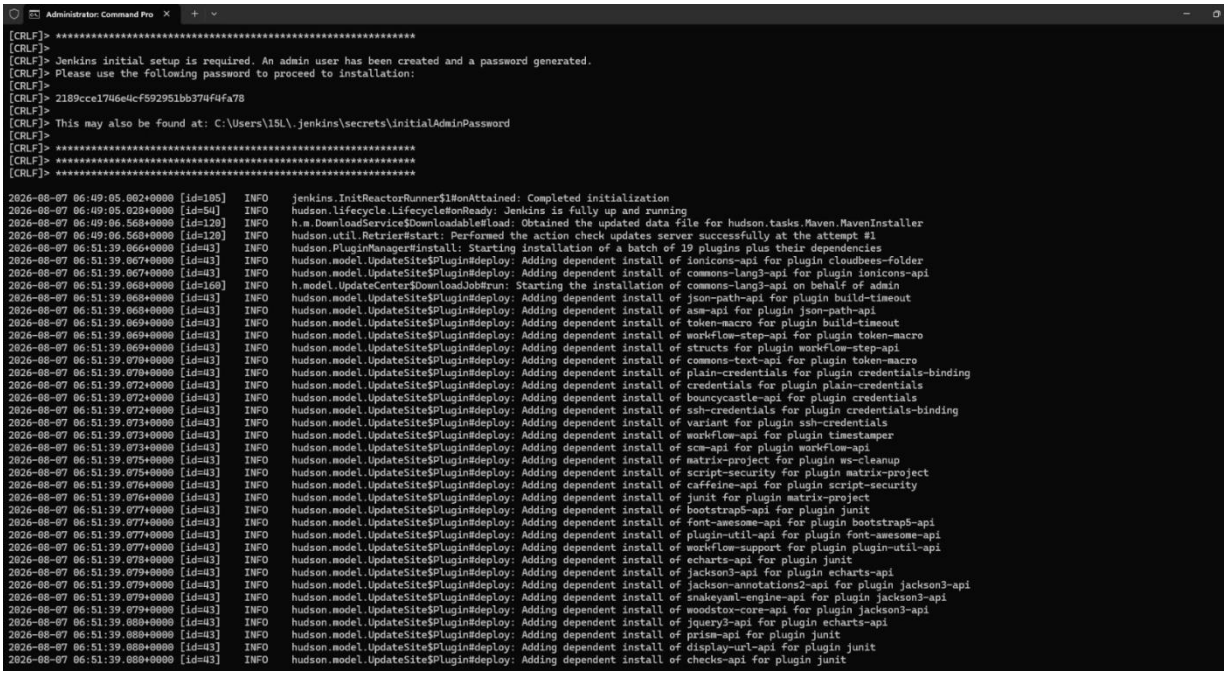
```
Administrator: Command Pro
[CRLF]> ************************************************************
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> 2189cce1746e4cf592951bb374f4fa78
[CRLF]>
[CRLF]> This may also be found at: C:\Users\15L\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************

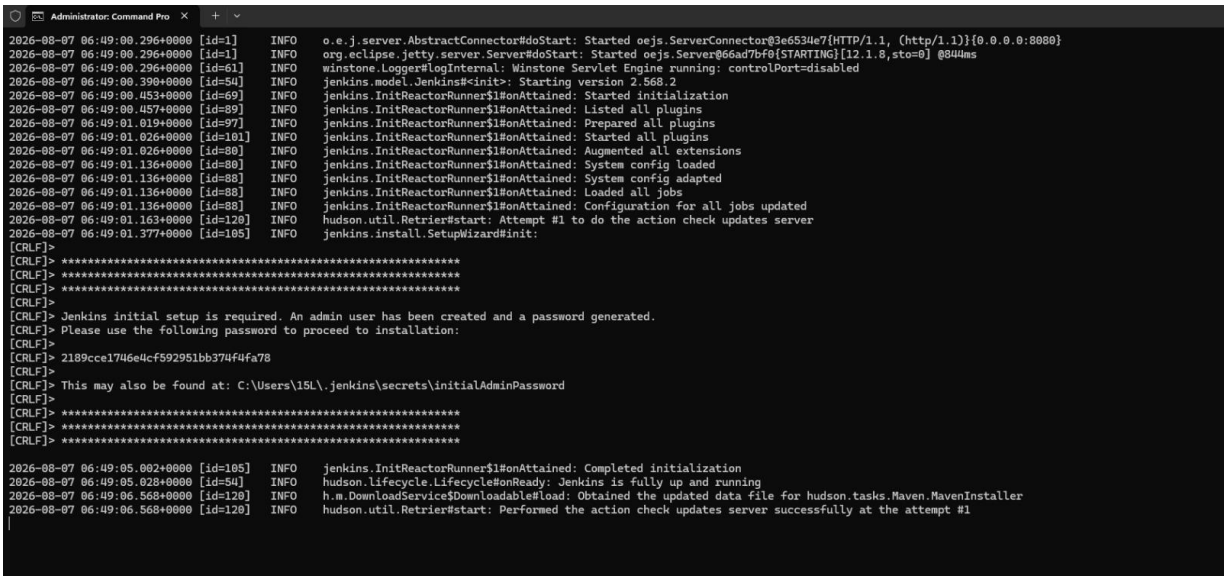
2026-08-07 06:49:05.002+0000 [id=105]   INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-07 06:49:05.028+0000 [id=54]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-08-07 06:49:06.568+0000 [id=120]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2026-08-07 06:49:06.568+0000 [id=120]   INFO    hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
2026-08-07 06:51:39.066+0000 [id=43]    INFO    hudson.PluginManager#install: Starting installation of a batch of 19 plugins plus their dependencies
2026-08-07 06:51:39.067+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of ionicons-api for plugin cloudbees-folder
2026-08-07 06:51:39.067+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of commons-lang3-api for plugin ionicons-api
2026-08-07 06:51:39.068+0000 [id=160]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of commons-lang3-api on behalf of admin
2026-08-07 06:51:39.068+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of json-path-api for plugin build-timeout
2026-08-07 06:51:39.068+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of asm-api for plugin json-path-api
2026-08-07 06:51:39.069+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of token-macro for plugin build-timeout
2026-08-07 06:51:39.069+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of workflow-step-api for plugin token-macro
2026-08-07 06:51:39.069+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of structs for plugin workflow-step-api
2026-08-07 06:51:39.070+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of commons-text-api for plugin token-macro
2026-08-07 06:51:39.070+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of plain-credentials for plugin credentials-binding
2026-08-07 06:51:39.072+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of credentials for plugin plain-credentials
2026-08-07 06:51:39.072+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of bouncycastle-api for plugin credentials
2026-08-07 06:51:39.072+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of ssh-credentials for plugin credentials-binding
2026-08-07 06:51:39.073+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of variant for plugin ssh-credentials
2026-08-07 06:51:39.073+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of workflow-api for plugin timestamper
2026-08-07 06:51:39.073+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of scm-api for plugin workflow-api
2026-08-07 06:51:39.075+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of matrix-project for plugin ws-cleanup
2026-08-07 06:51:39.075+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of script-security for plugin matrix-project
2026-08-07 06:51:39.076+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of caffeine-api for plugin script-security
2026-08-07 06:51:39.076+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of junit for plugin matrix-project
2026-08-07 06:51:39.077+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of bootstrap5-api for plugin junit
2026-08-07 06:51:39.077+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of font-awesome-api for plugin bootstrap5-api
2026-08-07 06:51:39.077+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of plugin-util-api for plugin font-awesome-api
2026-08-07 06:51:39.077+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of workflow-support for plugin plugin-util-api
2026-08-07 06:51:39.078+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of echarts-api for plugin junit
2026-08-07 06:51:39.079+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of jackson3-api for plugin echarts-api
2026-08-07 06:51:39.079+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of jackson-annotations2-api for plugin jackson3-api
2026-08-07 06:51:39.079+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of snakeyaml-engine-api for plugin jackson3-api
2026-08-07 06:51:39.079+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of woodstox-core-api for plugin jackson3-api
2026-08-07 06:51:39.080+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of jquery3-api for plugin echarts-api
2026-08-07 06:51:39.080+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of prism-api for plugin junit
2026-08-07 06:51:39.080+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of display-url-api for plugin junit
2026-08-07 06:51:39.080+0000 [id=43]    INFO    hudson.model.UpdateSite$Plugin#deploy: Adding dependent install of checks-api for plugin junit
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI				**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**

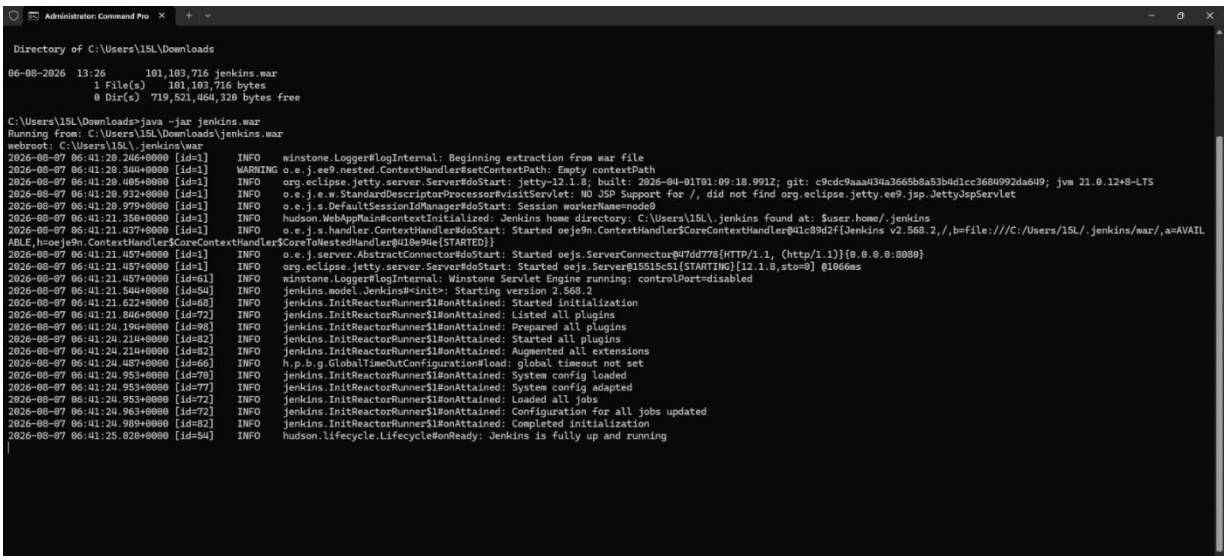


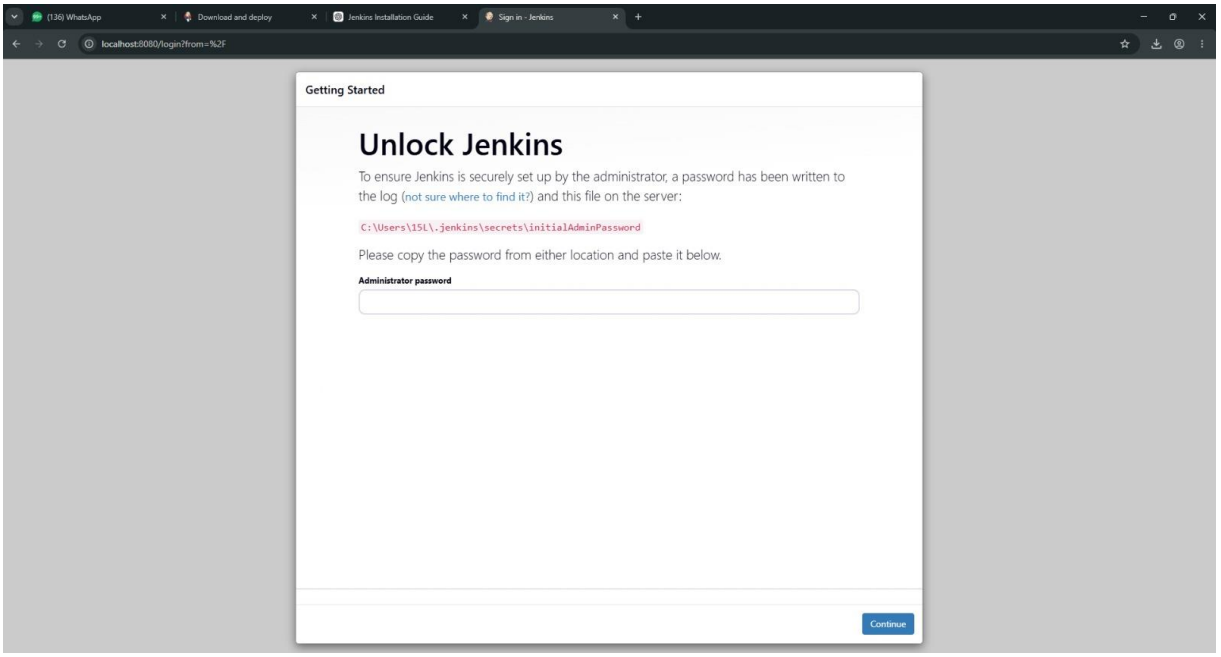


Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI				**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**

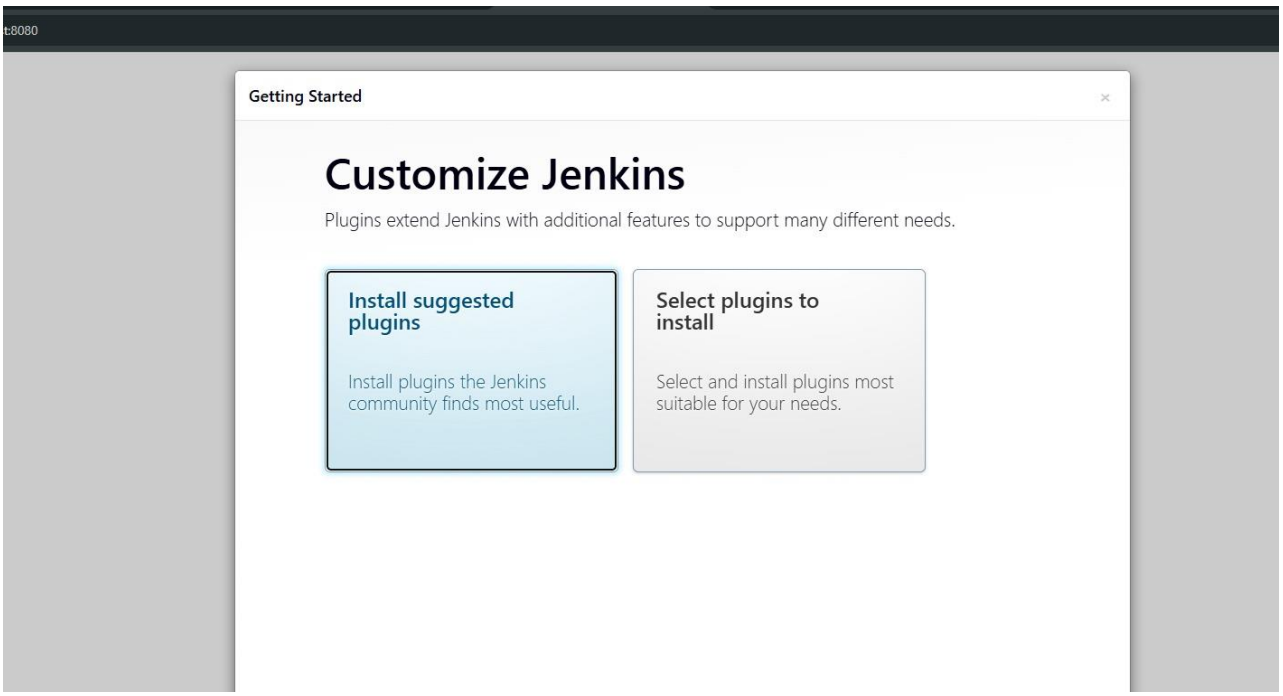


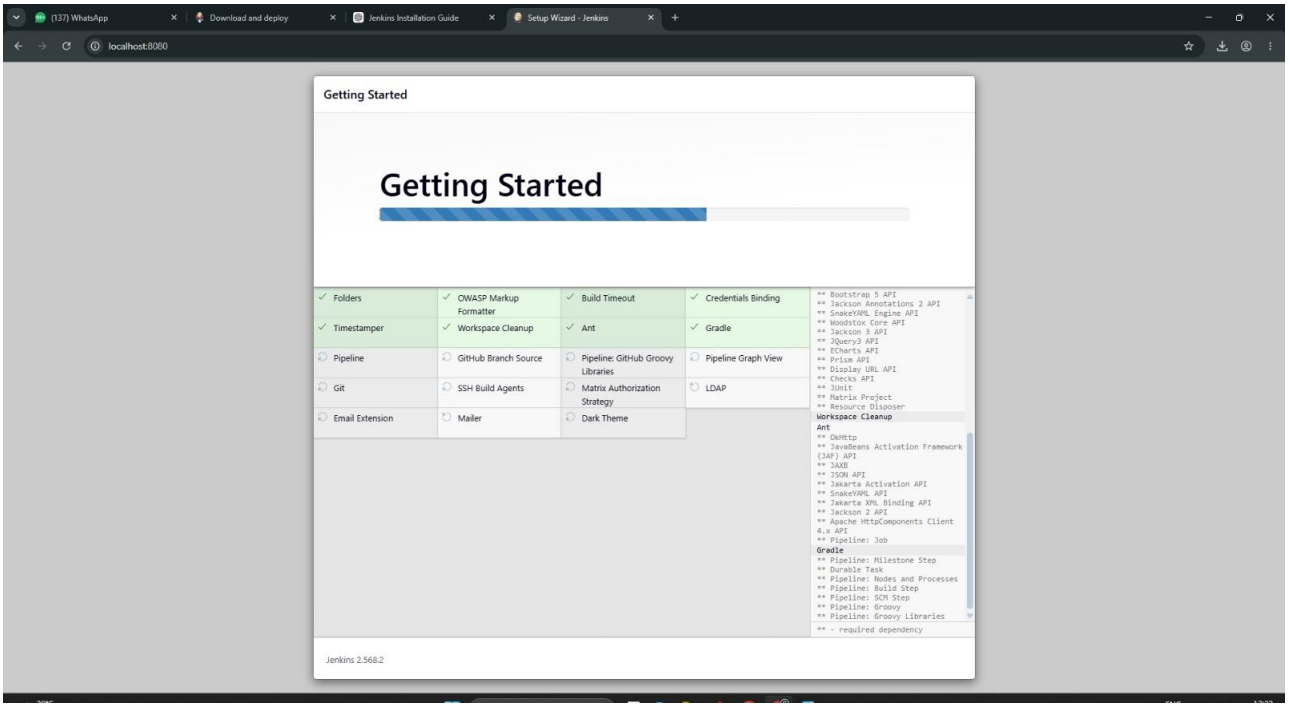


Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI			**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**

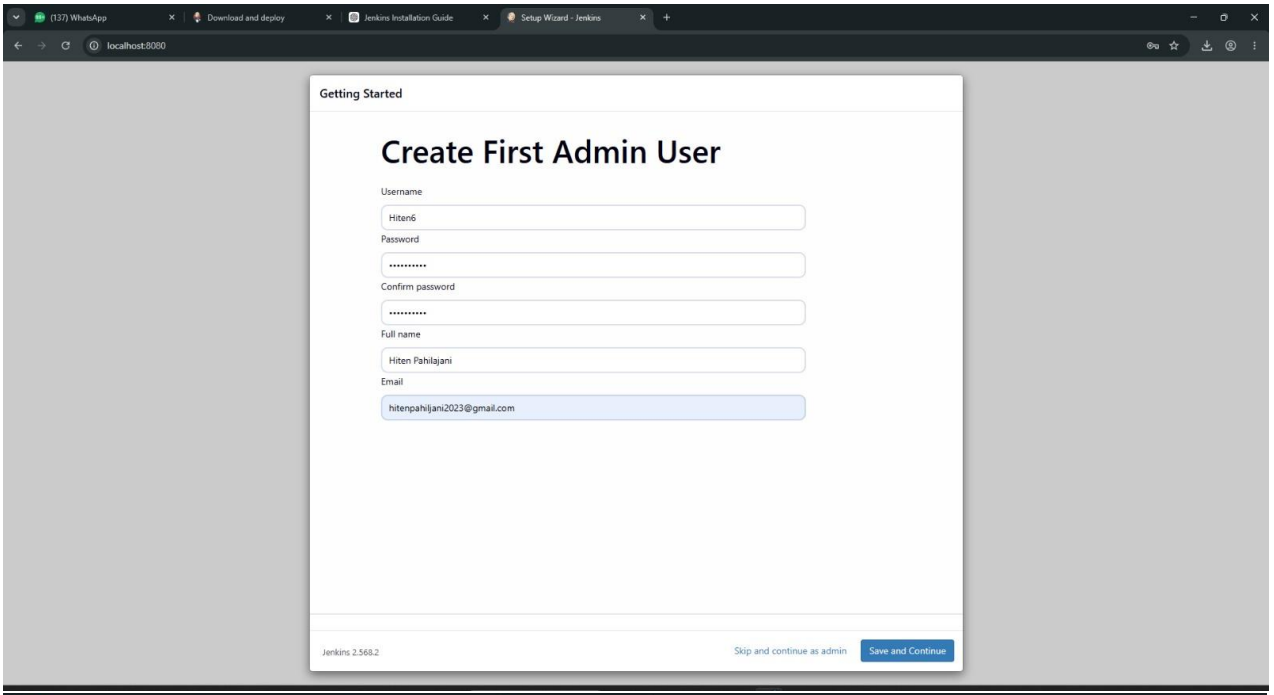


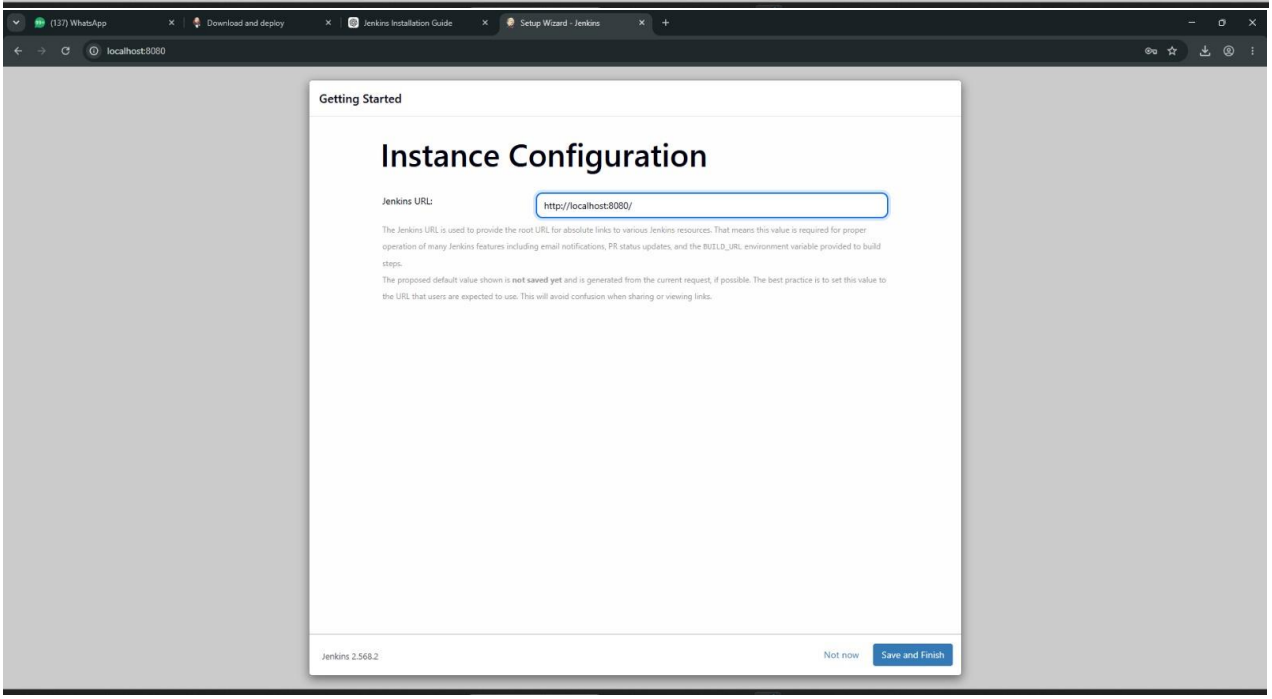


Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI			**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**



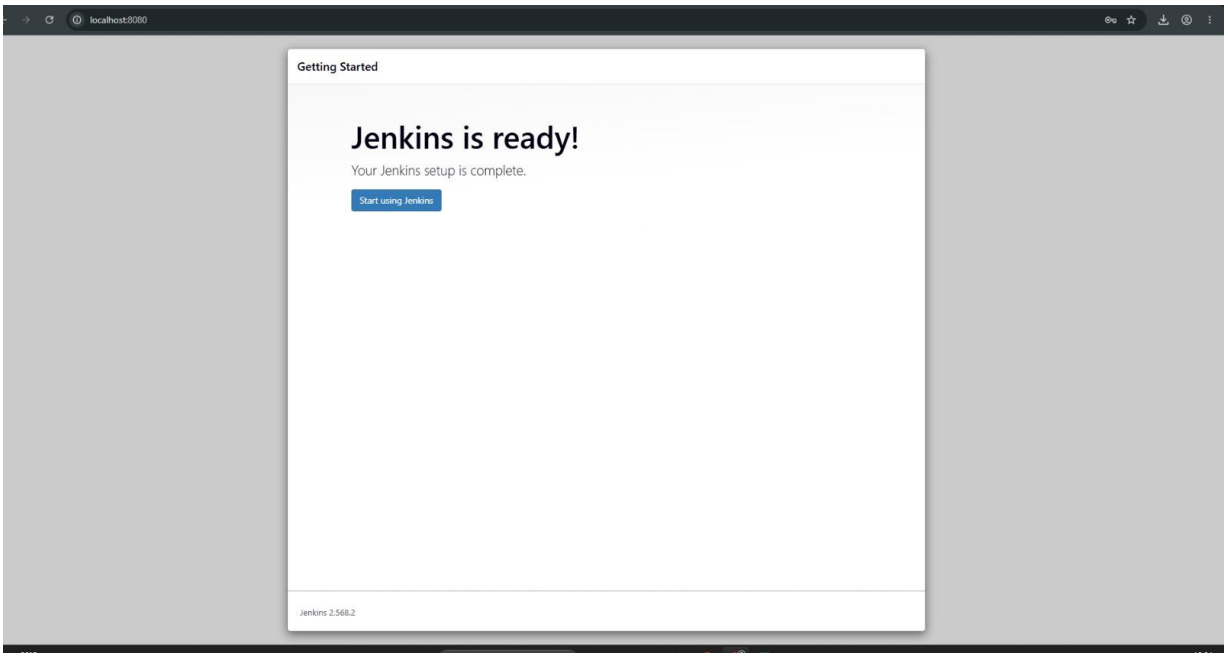


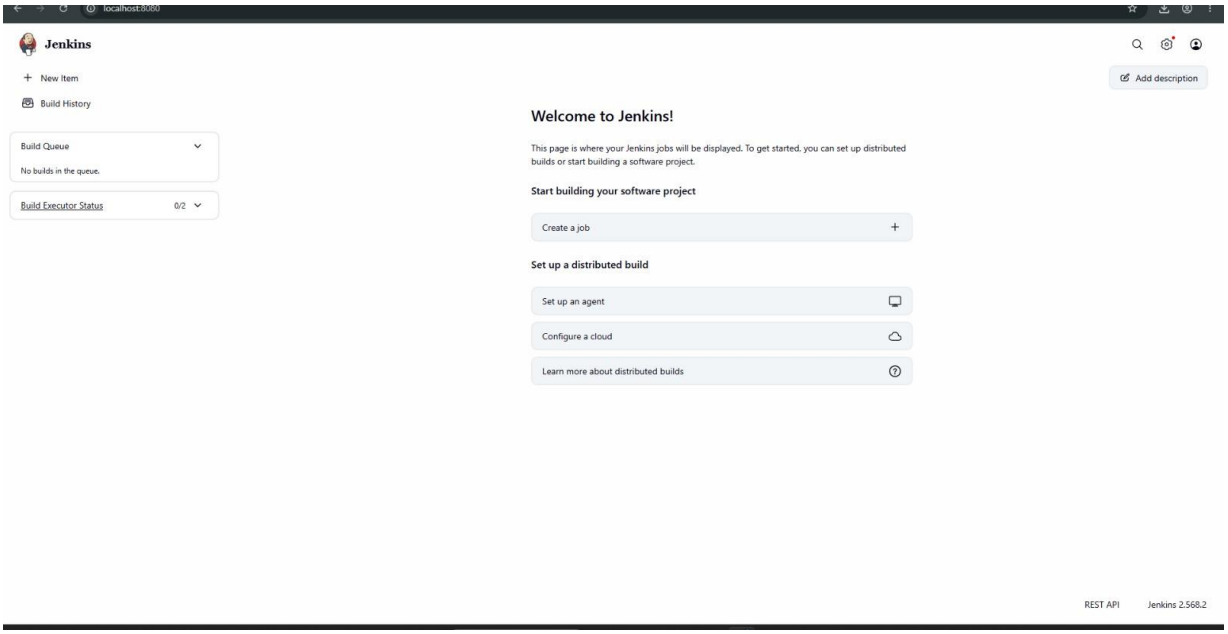
Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI			Subject: ASD&D
TE - 1	Batch: T13 Sem: 5	Roll No: 69	Date:07/08/2026

Step 4: Creating Pipelines

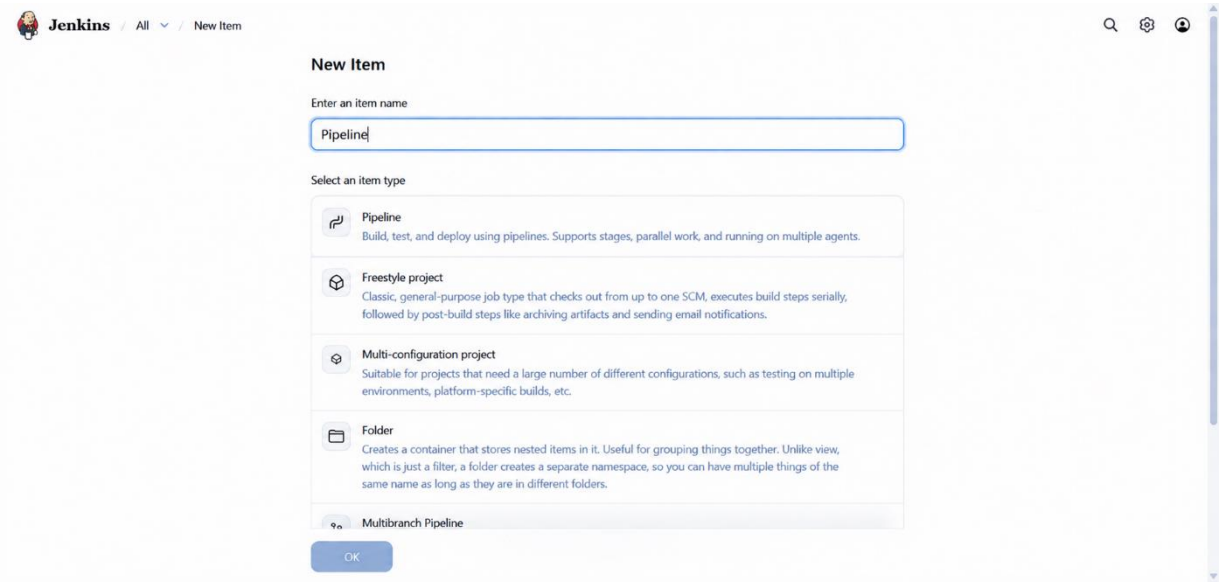




Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI			**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**



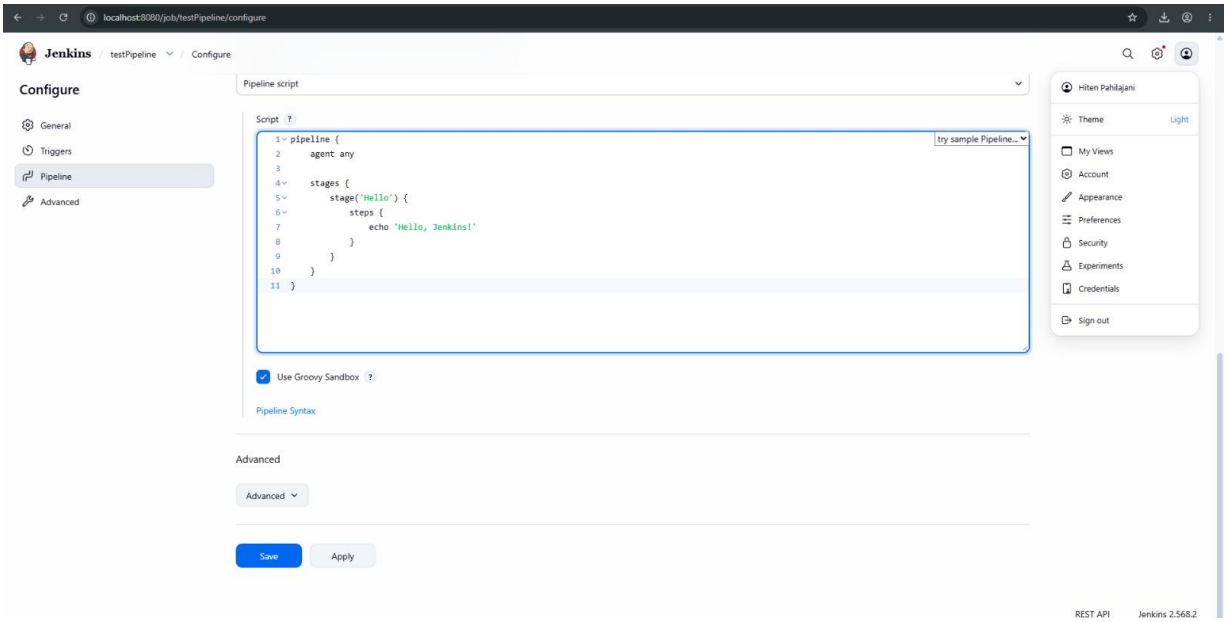
Code:

```
pipeline {    agent
any    stages {
stage('Hello') {
steps {
            echo 'Hello, Jenkins!!'
        }
    }
  }
}
```

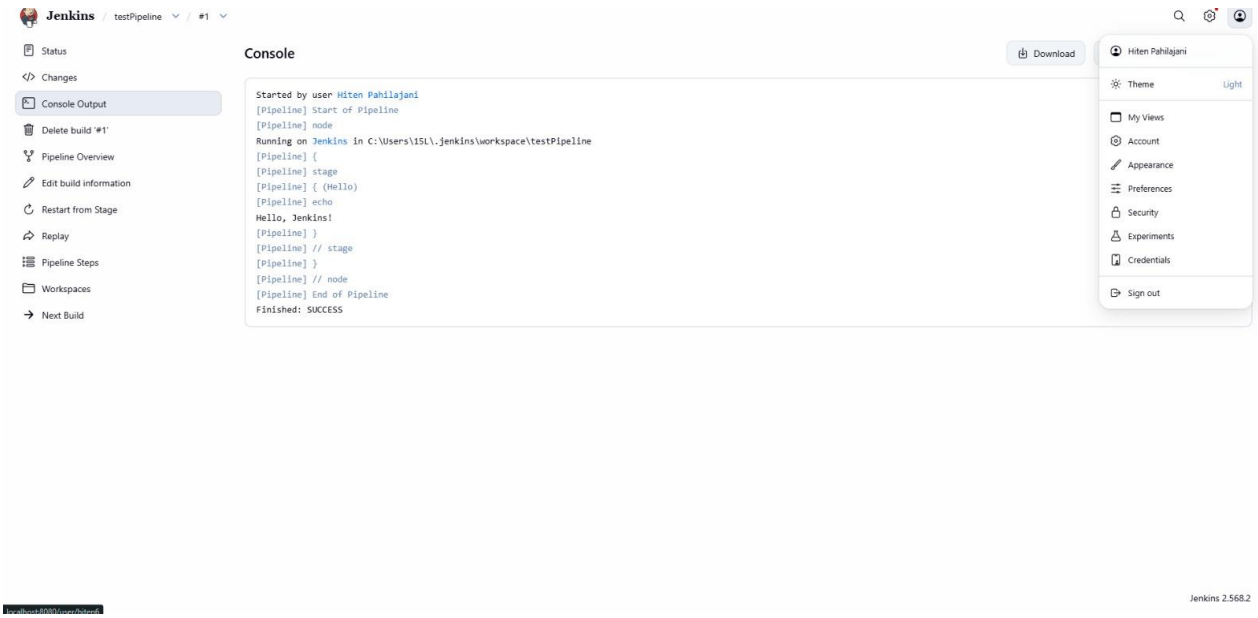
Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI			Subject: ASD&D
TE - 1	Batch: T13 Sem: 5	Roll No: 69	Date:07/08/2026



Output:

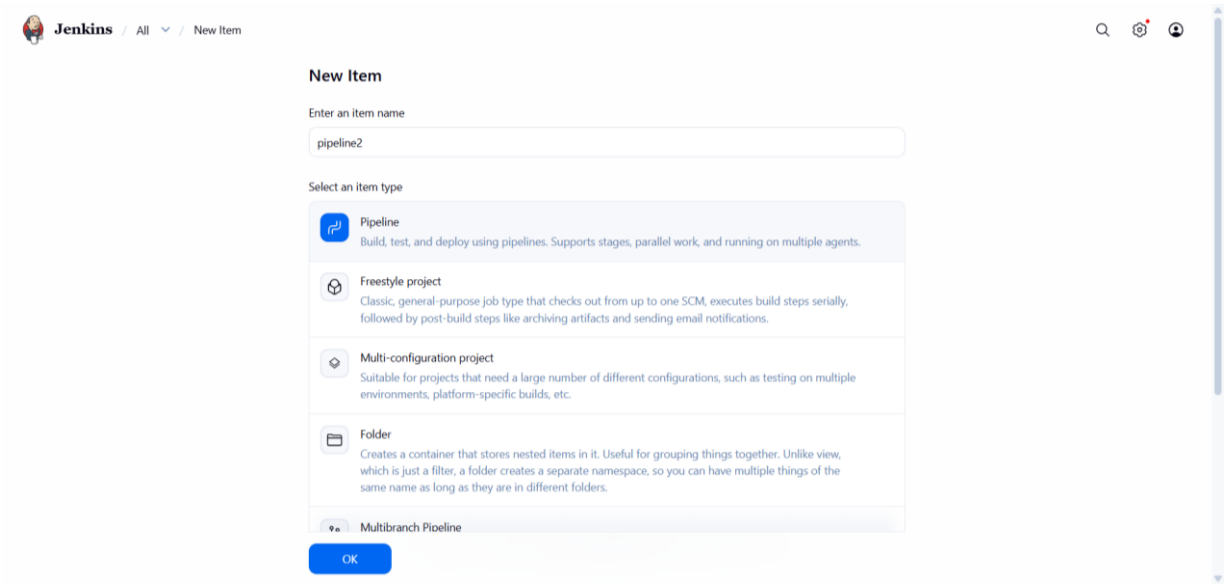


Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI				Subject: ASD&D
TE - 1	Batch: T13	Sem: 5	Roll No: 69	Date:07/08/2026

Step 5: Creating Scripted Pipeline



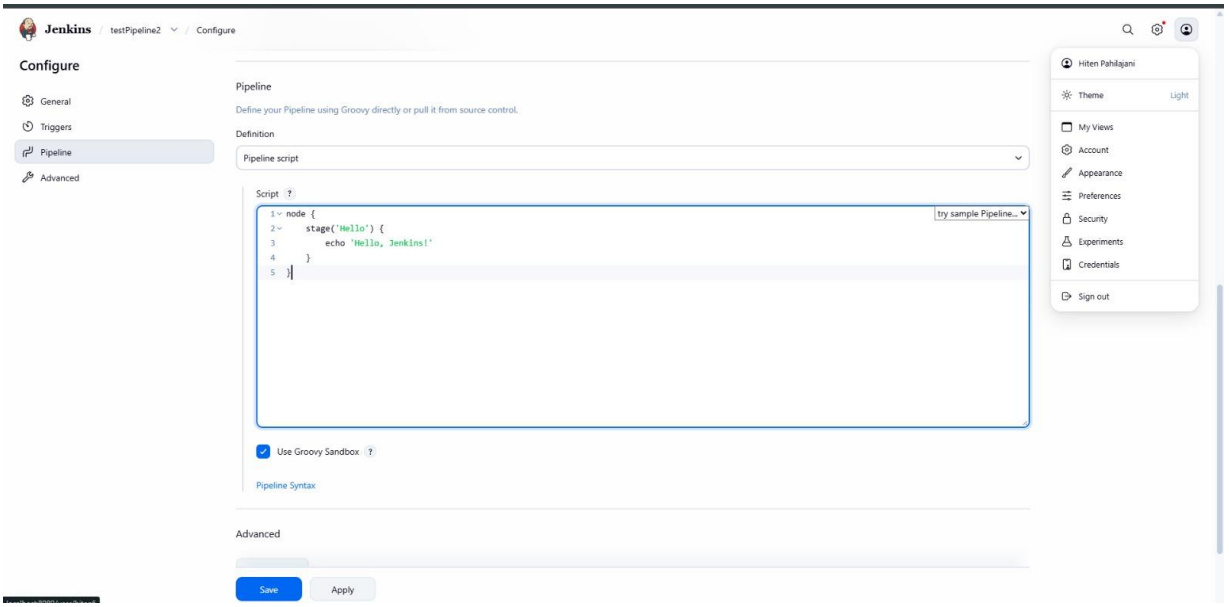
Code: node {

```
stage('Hello') {
        echo 'Hello, Scripted Jenkins Pipeline!'
    }
}
```

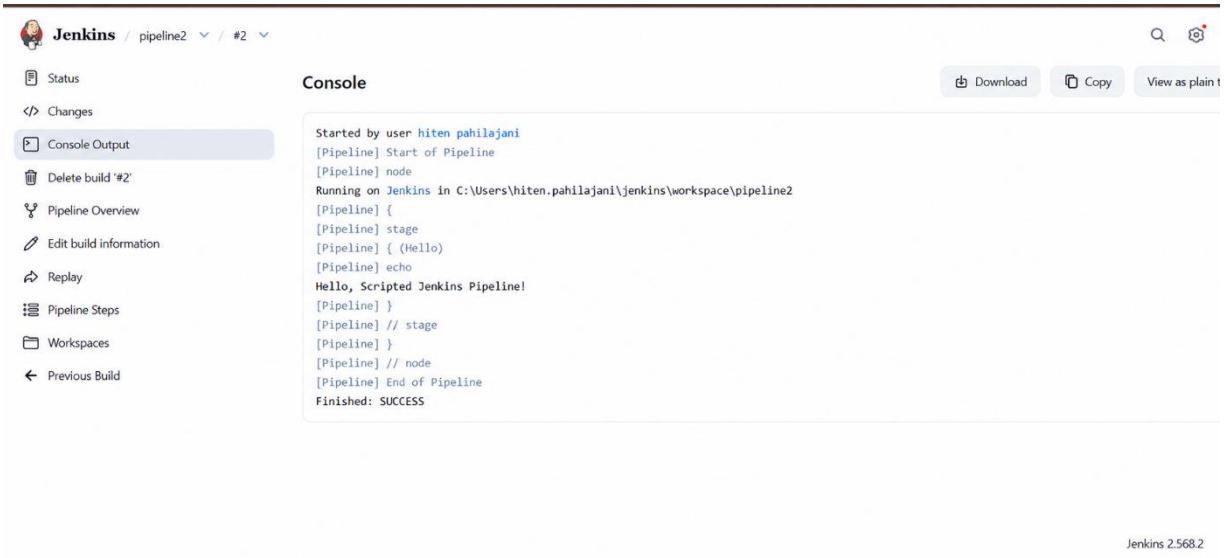
Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI			**Subject: ASD&D**
TE - 1	**Batch: T13**	**Sem: 5**	**Roll No: 69**	**Date:07/08/2026**



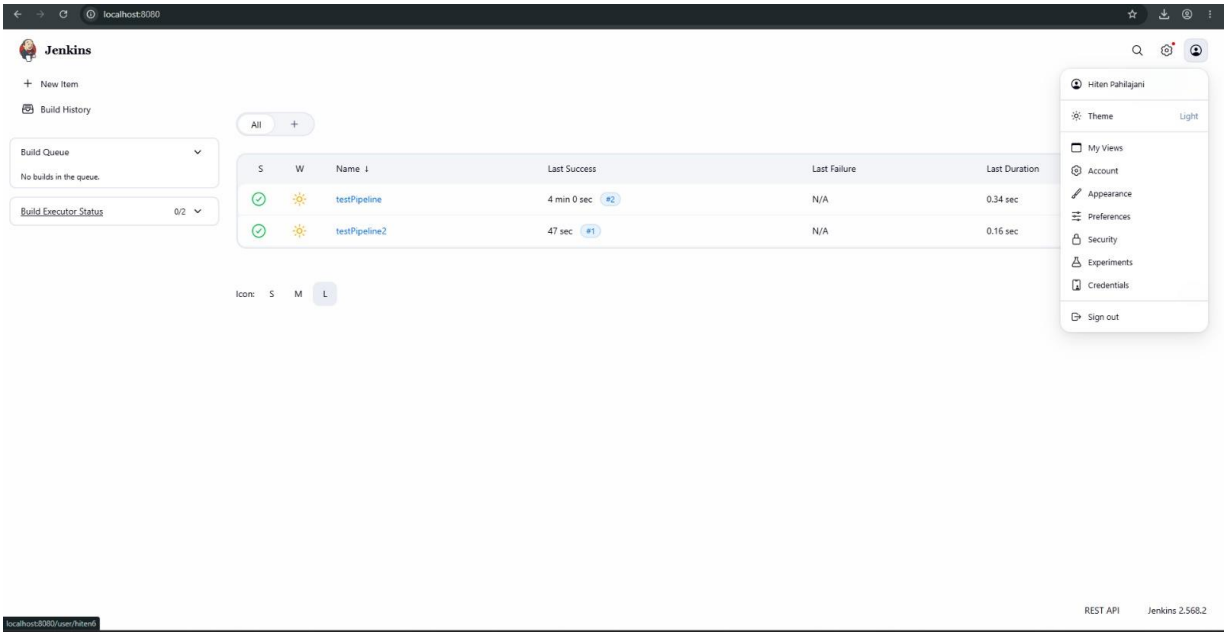
Output:



Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name:- HITEN PAHILAJANI			Subject: ASD&D
TE - 1	Batch: T13 Sem: 5	Roll No: 69	Date:07/08/2026



Conclusion:

Through this experiment, we learned the fundamentals of Continuous Integration (CI) and how Jenkins automates build and deployment workflows to eliminate manual integration errors. We gained hands-on experience verifying Java environments and deploying a standalone Jenkins server via the command-line interface. Additionally, we explored both Declarative and Scripted pipeline syntaxes, understanding how to structure automated build stages using code. By monitoring the console logs, we successfully executed and verified both pipeline builds. Overall, this experiment demonstrated how Jenkins serves as a crucial CI/CD automation tool.